
ML4SCIENCE-BENCH: Benchmarking Autonomous Agents on End-to-End Machine Learning for Science

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Abstract

Coding agents can write and execute code, but completing a scientific machine-learning task requires more than producing an executable program. An agent must construct a valid end-to-end workflow: prepare data correctly, prevent evaluation leakage, select appropriate models and evaluation procedures, consider domain-specific scientific constraints, and produce evidence that supports its conclusions. We introduce ML4SCIENCE-BENCH, a benchmark containing 100 executable tasks inspired by scientific papers across biology, chemistry, mathematics, and physics. Each task requires an agent to develop, analyze, or examine a complete machine-learning workflow in a terminal environment. Task-specific deterministic tests not only evaluate execution outcome, but also ML logic, scientific consistency, and reproducibility. We evaluate 10 language models, using a common TERMINUS-2 scaffold with all models configured for high reasoning effort. Despite access to the same environment and tools, the fraction of runs passing every test ranges from 24.7% to 77.3%. Average cost per run varies by more than sixfold, and higher success rates do not consistently correspond to higher cost. These results reveal a substantial gap between generating executable code and carrying out scientifically valid machine-learning workflows, and show that agent evaluation in scientific ML must jointly consider correctness, reliability, cost, and execution time.

1 Introduction

Machine learning for science couples data-driven models with simulators, mechanistic equations, surrogate models, and domain-specific software. A workflow can be numerically successful yet scientifically invalid: spatial or temporal leakage can inflate apparent accuracy, a surrogate can extrapolate beyond its domain of validity, a fitted parameter can violate unit consistency, or uncertainty can estimates can become severely miscalibrated under distribution shift [22, 45, 46, 48, 87, 90]. These failure modes make an end-to-end scientific ML a demanding test of autonomous agents. Success not only requires executable code, but also a defensible chain from data preparation and modeling assumptions to validation and auditable outputs.

Scientific machine-learning benchmarks such as PDEBench, Matbench, and OC20 are non-agentic: evaluate machine-learning methods on predefined scientific prediction and simulation-related tasks. They are not agent benchmarks and do not require an autonomous agent to design, execute, and validate an end-to-end scientific-ML workflow [15, 29, 102]. Current Agentic benchmarks test ML engineering, paper replication, reproducibility, scientific tool use, or discovery [14, 19, 74, 95, 97,

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101]. What remains under-tested is whether an agent can construct and validate sound machine-learning scientific workflow when consequential choices in data handling, modeling, and evaluation are left open.

ML4SCIENCE-BENCH targets this gap with 100 literature-motivated tasks. A task is included when it defines a scientific objective and makes learning or statistical inference central to the solution and requires multiple interdependent stages, such as preprocessing, data splitting, model fitting, validation, and generation of scientific artifacts within a controlled, sandboxed terminal environment. The agent receives task instructions together with local data and scientific resources, and must produce outputs that may include code, fitted models, numerical summaries, and machine-readable artifacts. Strict completion requires passing every task-specific deterministic test. Consequently, a strong predictive performance cannot compensate for invalid scientific quantities, flawed methodology, or internally inconsistent artifacts.

A concrete example. In the Alzheimer’s-inhibitor task, the agent receives molecular records, and an assay workbook, and must compare candidate inhibitors for BACE-1, AChE, and GSK-3B [4]. A successful workflow must derive activity from assay thresholds, reconcile records by semantic identity rather than row position, split BindingDB compounds reproducibly, perform feature selection using training data only, select a model for each molecular fingerprint, validate it externally on ChEMBL compounds, quantify drug-likeness, and integrate these signals into an admissible target recommendation. The verifier independently recomputes key quantities and checks the resulting artifacts for methodological and scientific consistency.

In this work, our contributions are:

- A benchmark of 100 cross-domain tasks that makes construction and audit of the complete scientific-ML workflow, rather than final prediction alone, the evaluation object;
- A deterministic evaluation contract that tests execution, ML methodology, scientific validity, and reproducibility, with strict completion requiring every task-specific test to pass;
- A controlled 3,000-run evaluation of 10 LLMs, including task-bootstrap uncertainty, task and domain-level analyses, and comparisons across capability, latency, and cost;

2 Related benchmarks and positioning

Scientific-ML benchmarks such as PDEBench, Matbench, and OC20 are non-agent benchmarks³[15, 29, 102]. MLE-bench evaluates autonomous ML engineering on Kaggle competitions [14]; PaperBench evaluates from-scratch replication of AI research with author-developed rubrics [101]; and CORE-Bench tests the computational reproducibility of existing scientific repositories [97]. ScienceAgentBench and SciAgentGym broaden the focus to data-driven scientific programming and multi-step tool use [19, 95]. BioXArena is the closest end-to-end scientific-ML analog, but specializes in biomedical prediction with hidden labels and biologically aware metrics [58].

Table 1 makes the intended intersection explicit. First, a learned pipeline is central: the agent must fit, select, calibrate, emulate, invert, or otherwise validate a learned component. Second, the task exposes scientific structure. Third, success is defined at the workflow level. A run may execute successfully and achieve a plausible predictive score yet still fail because of leakage, hidden-test refitting, invalid units, a violated scientific invariant, or missing required scientific artifact. ML4SCIENCE-BENCH therefore complements prediction benchmarks, replication benchmarks, and scientific tool-use benchmarks. Appendix F discusses additional benchmarks including MLGym, MLAgentBench, AIRS-Bench, FML-bench, AstaBench, MDGym, COMPOSITE-STEM, Discovery-Bench, and NewtonBench [11, 43, 55, 70, 74, 84, 110, 122, 127].

3 Benchmark design

Operational scope. Each task places the agent in a sandboxed terminal workspace containing instructions, local datasets, and scientific or machine learning libraries required for completion. Outputs may include executable code, fitted models, predictions, numerical summaries, figures, or machine-readable artifacts. Tasks are included only when they center on a scientific objective requiring

³The model is evaluated through direct API calls without an agent harness.

Table 1: Comparison with representative adjacent benchmarks. “Process validity” means that evaluation checks consequential workflow decisions, for example split leakage, model-selection discipline, units, or scientific constraints, rather than relying only on execution or a final task metric. “E2E ML” means that the agent must construct or fit a learned pipeline. “Yes” means the property is required throughout the benchmark; “no” means it is absent from the benchmark-level contract; and “partial” means that some consequential workflow choices are checked but not all of the process-validity categories used here. “Rubric” denotes process evidence assessed through a LLM judged rubric rather than deterministic process tests, while “fixed” means the relevant workflow choices are supplied by the benchmark and therefore are not evaluation targets. “No agent” denotes a predictive benchmark in which an autonomous agent does not construct the workflow. These are descriptive categories, not an ordinal quality score. “Mixed” indicates that the property applies to a subset of tasks. Entries describe benchmark-level evaluation contracts; individual tasks may provide additional tests.

Benchmark	Scope	E2E ML	Process validity	Scientific coverage	Primary evaluation signal
MLE-bench [14]	75 Kaggle competitions	yes	no	general ML	private competition metric and medal threshold
PaperBench [101]	20 AI-paper replications	yes	rubric	AI/ML research	author-developed hierarchical rubric with an LLM judge
ScienceAgent-Bench [19]	102 paper-derived tasks	mixed	partial	four scientific disciplines	program, execution result, output, and cost
CORE-Bench [97]	270 tasks from 90 papers	no	no	cross-science	repository execution and reproduction-answer correctness
SciAgentGym [95]	259 tiered tool-use tasks	mixed	partial	four natural sciences	task success and tool-path efficiency
BioXArena [58]	76 biomedical ML tasks	yes	partial	nine biomedical domains	hidden-label prediction with biology-aware metrics
PDEBench / Mat-bench / OC20 [15, 29, 102]	fixed scientific-ML tasks	no agent	fixed	physics, materials, catalysis	task-specific predictive or physics-aware metric
ML4SCIENCE-BENCH	100 tasks based on scientific papers for ML4Science	yes	yes	cross-science (biology, chemistry, mathematics, and physics)	strict task-specific tests of execution, ML logic, and scientific outputs using deterministic tests

multi-step statistical inference or learning. Pure question-answering, deterministic simulations lacking a learned component, and tasks reducible to a single API call fall outside this benchmark’s scope. All tasks are packaged using the Harbor framework [38], which standardizes instructions, supporting files, one solution, pytest, and environment specifications. All tasks are CPU-friendly.

Coverage and provenance. The evaluated benchmark version comprises 44 biology tasks, 13 chemistry tasks, 8 mathematics tasks, and 35 physics tasks. Tasks span surrogate modeling, inverse problems, calibration, structured validation, equation discovery, causal analysis, molecular and materials modeling, biological prediction, and scientific audit. Most use synthetic dataset, supporting controlled task construction and held-out evaluation; a smaller subset uses data borrowed from the motivating publication. Appendix Table 9 lists all evaluated task identifiers, provenance labels, outcomes, and citations.

Literature-to-task curation. A candidate task must satisfy three inclusion criteria. First, it must be anchored in a specific primary publication with verifiable scientific ground truth or outcome. Second, it must contain a computationally substantive core that can be reconstructed as a self-contained problem. Third, the task also must require multiple interdependent implementation and validation decisions by domain experts that could not be completed through a single API call. The motivating publication defines the scientific setting and methodological concept, but each benchmark task is independently constructed. Domain experts identify a derivation, model, or workflow to reconstruct, then translate it into an original executable task with new inputs, required artifacts, and deterministic tests that preserve the scientific constraints of the source problem without reproducing the original experiment verbatim.

Pre-evaluation construction and quality assurance. Before evaluation, we followed a three-stage quality-control process. First, domain experts created an initial pool of 200 tasks inspired by scien-

tific papers, with 50 tasks from each scientific subject. Second, every strict test was mapped to a corresponding requirement in the task instructions or environment specification. Finally, independent expert reviewers examined tasks for unclear instructions, missing dependencies, environment problems, incorrect tests, and other task-design issues. Tasks that did not meet the required quality standards were removed. This process produced a final set of 100 tasks, which was fixed before evaluation and used for benchmark evaluation without further changes. These steps reduce the likelihood that benchmark defects contribute to observed failures and improve the overall quality and reliability of the task set.

Task Instruction (Contract). Each task’s instruction specifies a scientific question, inputs, required outputs, and executable entry point. Its learning contract determines what must be fitted, selected, calibrated, inferred, or tested. The agent is responsible for developing the workflow rather than filling a fixed model template. This design makes choices such as preprocessing scope, validation structure, baseline construction, uncertainty estimation, and scientific diagnostics observable to the verifier.

Verifier. Task-specific tests target four complementary layers: (i) execution and required artifacts; (ii) ML methodology, including leakage prevention and model-selection discipline; (iii) scientific validity, including units, ranges, invariants, calibration, and task-specific constraints; and (iv) reproducibility in the task environment. Let $z_{tmrk} \in \{0, 1\}$ denote the result of test k for task t , model m , and repetition r . We define strict completion as

$$y_{tmr} = \prod_{k=1}^{K_t} z_{tmrk}, \quad (1)$$

where K_t denotes the number of tests recorded for the task t . Thus every required test must pass; strong performance on one output cannot compensate for a fatal workflow error. The verification tests are never mounted into the agent workspace; they are injected only at verifying time.

Solution (Oracle). Each task includes a complete golden certified solution prepared by a domain expert. The golden certified solution shows that the task can be completed using the provided data, software, and computing resources. It also produces the expected results and artifacts used to check the verification tests. The agent does not have access to this solution or its outputs in its environment. Before a task is included in the evaluation, the golden certified solution is run to confirm that the required resources are available, and the tests behave as intended. All solutions were successfully reproduced by the authors in the controlled benchmark environment.

4 Evaluation protocol

Models and repetitions. We evaluate ten LLMs using the same TERMINUS-2 scaffold: Claude Opus 4.8, Claude Opus 5, Claude Sonnet 5, GPT-5.6 Sol, GPT-5.6 Terra, Gemini 3.1 Pro Preview, Gemini 3.7 Flash, Grok 4.6, Kimi K3, and GLM 5.2. Every backend is evaluated in high-thinking mode⁴; no low- or medium-thinking results are mixed into the comparison. Every model receives three independent attempts on every task, giving 300 runs per model and 3,000 overall. Holding the scaffold and thinking mode fixed isolates the effect of the scaffolding as far as the common interface permits; the experiment is not a comparison of each provider’s best native agent product. Agents had access to the internet during evaluation. The task instructions, inputs, and required outputs were newly created rather than copied directly from the motivating papers, reducing the possibility of exact task memorization. However, agents could retrieve related publications. The hidden verification tests, golden certified solutions, and expected outputs were not accessible to the agents. Each task includes a `task.toml` file that records the container image used to create its sandboxed environment and the settings needed to build, run, and evaluate the task. These settings specify the time limits for preparing the environment, running the agent, and checking its outputs; and the available CPU, memory, and storage. The settings may vary according to the needs of each task, but every model receives the same configuration for a given task.

⁴For each model, we selected the high reasoning setting available through its API. This asks the provider to allocate a high reasoning budget to the model. However, providers may define and implement this setting differently; therefore, “high” does not imply that all models use the same number of reasoning tokens or the same amount of computation.

Table 2: TERMINUS-2 results over 100 tasks and three runs per task, with every model evaluated in high-thinking mode. Strict pass requires all task tests to pass. Continuous reward is the run-normalized fraction of tests passed in Eq. 3. Confidence intervals are percentile intervals from 20,000 task-level bootstrap resamples. Cost and execution time are run-weighted means.

Model	Passes	Pass rate	95% CI	Continuous reward	Time (min)	Cost (\$)	\$/pass
Opus 5	232	77.3	[70.3, 84.0]	0.984	17.1	1.99	2.58
GPT-5.6 Sol	219	73.0	[65.7, 80.0]	0.967	7.2	0.91	1.25
Grok 4.6	216	72.0	[65.0, 78.7]	0.969	28.9	1.19	1.66
Opus 4.8	194	64.7	[56.7, 72.7]	0.962	13.4	1.94	3.01
Sonnet 5	169	56.3	[48.7, 64.0]	0.953	22.2	1.66	2.95
GPT-5.6 Terra	167	55.7	[47.3, 64.0]	0.943	5.9	0.32	0.58
Gemini 3.7 Flash	145	48.3	[41.0, 55.7]	0.789	10.5	0.99	2.04
GLM 5.2	140	46.7	[38.7, 54.7]	0.924	21.9	0.58	1.24
Kimi K3	114	38.0	[30.3, 45.7]	0.913	19.7	0.68	1.79
Gemini 3.1 Pro	74	24.7	[18.0, 32.0]	0.821	8.6	0.79	3.20

Metrics. Let $p_{tm} \in \{0, 1, 2, 3\}$ be the number of strict passes for task t and model m . The primary score is the task-macro strict pass rate

$$P_m = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{p_{tm}}{3}, \quad (2)$$

where $\mathcal{T}$ is the set of evaluated tasks. Because every task has three runs per model, this equals the micro-average over runs.

To retain information below the all-or-nothing threshold, let $f_{tmr} = \sum_{k=1}^{K_t} (1 - z_{tmrk})$ be the number that failed. We define the run-level continuous reward and its model-level average as

$$c_{tmr} = 1 - \frac{f_{tmr}}{K_t} = \frac{1}{K_t} \sum_{k=1}^{K_t} z_{tmrk}, \quad C_m = \frac{1}{3|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{r=1}^3 c_{tmr}. \quad (3)$$

Thus each run and task receive equal weight, while tests are weighted equally within a task. A strict pass has $c_{tmr} = 1$; a failed run retains the fraction of tests it passed. We report C_m as a descriptive secondary metric rather than a substitute for strict pass because tests differ in granularity and scientific consequence. More details are provided in Appendix B.

Cost per strict pass is total model-call cost divided by the number of strict passes; it excludes host-compute and storage costs.

Operational definitions. A *trajectory step* is one top-level entry in the TERMINUS-2 trajectory, including the initial task message and each subsequent agent episode; one step can contain zero, one, or several tool calls. A *tool call* is one structured invocation recorded in a step’s `tool_calls` array, so a single call can itself submit multiple shell commands. Wall-clock time is `agent_execution.finished_at` minus `agent_execution.started_at`. It includes model/API latency and terminal computation or waiting performed during the agent trajectory, but excludes environment provisioning, agent setup, artifact collection, and verifier/grading time.

Statistical analysis. We form 95% percentile-bootstrap confidence intervals by resampling the 100 tasks with replacement 20,000 times, using random seed 20260824. For pairwise comparisons, we resample the within-task score differences, thereby preserving the shared task set and the three attempts used to compute each task-level score. These intervals quantify sensitivity to the composition of the benchmark task set. They do not account for future model-version drift or all sources of inference-time variability, and three repetitions provide only a coarse characterization of within-model stochasticity.

5 Results

Leaderboard. Across all models, 1,670 of 3,000 runs pass strictly (55.7%). Table 2 reports the complete leaderboard. Opus 5 has the highest observed pass rate at 77.3%, followed by GPT-5.6 Sol

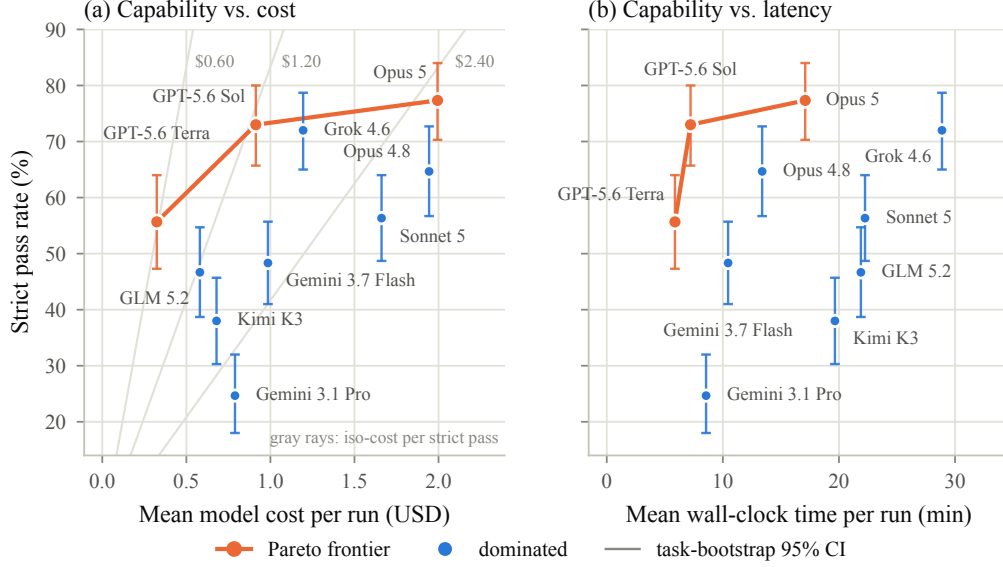


Figure 1: Capability against the two resources that buy it, under the common TERMINUS-2 high-thinking configuration. Vertical bars are the task-bootstrap 95% confidence intervals of Table 2; orange marks the observed Pareto frontier and gray rays in (a) are iso-cost-per-strict-pass contours. Up and to the left is preferable. The same three models, GPT-5.6 Terra, GPT-5.6 Sol, and Opus 5, are nondominated on both resources, and the intervals overlap widely among the leaders, so neither panel resolves a single best model.

at 73.0% and Grok 4.6 at 72.0%; the weakest model reaches 24.7%. The resulting 52.6-point range shows substantial sensitivity to the model backend under a fixed scaffold. More discussion regarding the leaderboard is provided in Appendix C.

Capability, cost, and latency give different orderings. GPT-5.6 Sol averages \$0.91 and 7.2 minutes per run, versus \$1.99 and 17.1 minutes for Opus 5, while trailing by 4.3 points. GPT-5.6 Terra is the cheapest model at \$0.32 per run and \$0.58 per strict pass, but its 55.7% pass rate is 17.3 points below Sol. Figure 1 makes the frontier explicit: the same three models, Terra, Sol, and Opus 5, are nondominated for both cost and wall-clock time, whereas Grok 4.6 buys its 72.0% with the longest runs in the study (28.9 minutes). Quality, latency, and cost should therefore be reported jointly rather than collapsed into one leaderboard. The campaign consumed 776 agent-execution hours and \$3,320 in model-call cost, of which \$1,519 (46%) and 376 hours bought no strict pass; failed runs are not cheap early exits in aggregate, averaging \$1.14 and 17.0 minutes against \$1.08 and 14.4 minutes for passing runs.

Difficulty spans the full range and is only partly shared. Figure 2(a) shows the outcome for every task–model pair. Four tasks receive no strict passes across the 30 attempts, whereas one task is passed in all 30 attempts (Appendix Table 7). Appendix A descriptively audits the top-three-model failures on those four tasks. Pooling results across models shows that most of the benchmark is empirically attainable: 96 of the 100 tasks are passed at least once by at least one model, and 85 are passed in all three attempts by at least one model.

Mixed outcomes. Let θ_{tm} be the per-attempt success probability for task t and model m . Under conditionally independent Bernoulli attempts, the probability of one or two successes in three trials is

$$\Pr(1 \leq p_{tm} \leq 2 \mid \theta_{tm}) = 3\theta_{tm}(1 - \theta_{tm}), \quad (4)$$

which is maximized at 0.75 when $\theta_{tm} = 0.5$. The mixed share therefore depends intrinsically on capability and task difficulty as well as any additional run-to-run dependence. We retain it only as a descriptive finite-repetition diagnostic: 333 of the 1,000 task–model cells are mixed, spanning 92 tasks. Opus 5 passes 88 tasks at least once and 64 on all three; Gemini 3.7 Flash passes 73 at least once and 22 on all three. Scoring by “solved at least once in three” reorders the table: Grok 4.6 (89)

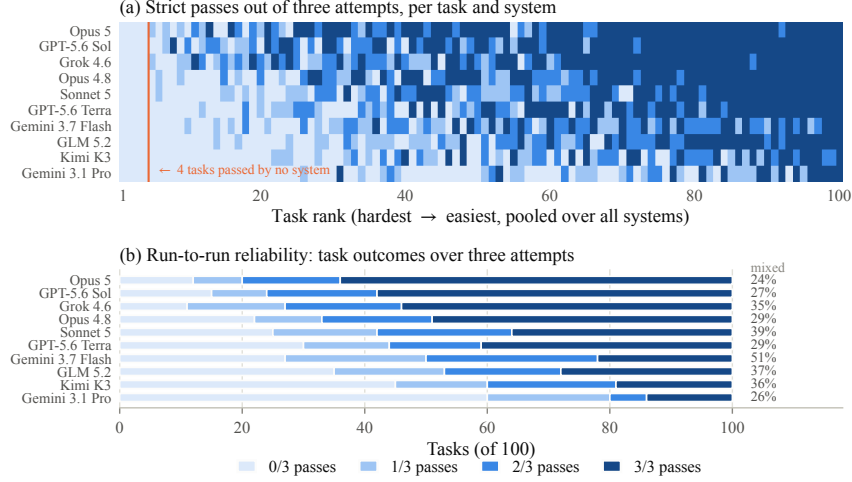


Figure 2: Where difficulty lives, and how repeatable a strict pass is. (a) Every task–model cell, shaded by strict passes out of three attempts of each model (system); tasks are ordered by pooled pass count over all 30 attempts and models are ordered by strict pass rate. (b) The same outcomes summarized per model: a task is *mixed* when it is passed on one or two of three attempts.

edges past Opus 5 (88), and Flash (73) overtakes GPT-5.6 Terra (70) despite trailing it by 7.4 points on strict pass rate.

Interaction volume is confounded by task difficulty and agent policy. Interaction volume (i.e. number of steps and tool calls) varies substantially despite the shared scaffold (Appendix Table 6 and Figure 4): Gemini 3.7 Flash uses the most trajectory steps (56.9 per run) and Sonnet 5 the most tool calls (53.2 per run), while GPT-5.6 Sol averages 12.3 steps per run and needs only 16.9 steps and 33.3 calls per strict pass, against 26.5 and 35.4 for Opus 5. Calls per step range from 0.76 (Flash) to 2.05 (GLM 5.2), so a trajectory step is not a model-independent unit of tool activity. Interaction volume is therefore not a pure measure of either the task difficulty or agent policy. Across tasks (Figure 3), pass rate correlates negatively with mean steps ($\rho = -0.51$) and tool calls ($\rho = -0.46$), largely because difficult tasks elicit longer trajectories; after removing additive task and model means the cell-level correlations shrink to $r = 0.03$ and $r = -0.02$. The comparison even changes sign once task and model are held fixed. Pooled over all 3,000 runs, failing attempts are the *longer* ones, by 4.5 steps and 7.5 calls, because failures concentrate on hard tasks and on models with more verbose interaction policies; within the 333 mixed cells the ordering reverses, with passing attempts using 5.1 additional steps (95% CI [3.3, 7.0]) and 3.2 additional calls (95% CI [1.4, 5.1]). The task-level association and the large between-model differences in Table 6 show that raw interaction counts are jointly confounded by task difficulty and agent policy.

Subject-level differences. Subject-level scores (Appendix Table 5 and Figure 5(a)) differ systematically. Mathematics has the highest pass rate for every model, although it is also the smallest group at 8 tasks; the relative ordering of biology, chemistry, and physics varies by model. Pass rates also differ by data provenance: the 12 tasks whose data matches the motivating publication pass at 48.1% against 57.6% for the 88 synthetic tasks. Both groupings are unbalanced and sometimes interdisciplinary, so these gaps can reflect task construction, verifier strictness, or subject-assignment conventions as well as model capability, and we do not interpret them as intrinsic domain difficulty.

Discussion: The results support following conclusions. First, end-to-end scientific ML remains imperfect under the evaluated scaffold: even the strongest observed model fails 68 of 300 runs, and no model clears every task. Second, model choice changes both capability and efficiency substantially; the strongest cost-quality operating point need not be the highest raw pass rate.

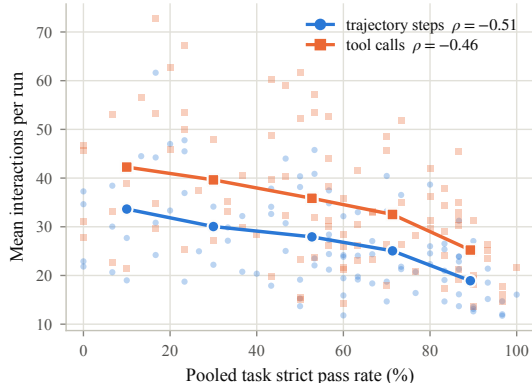


Figure 3: Interaction volume is associated with task difficulty and is also affected by agent policy. Each faint mark is one task averaged over all ten systems and three attempts; the connected marks are means within five equal-width pass-rate bins. Both trajectory length and tool use fall monotonically across the binned difficulty spectrum. This is a descriptive association, not evidence that reducing interaction would improve success.

6 Limitations

Limitations of the current evidence. The benchmark evaluates one scaffold and one high-thinking setting, so conclusions do not generalize to native agent products, lower thinking budgets, other resource limits, or future model snapshots. “High” is the selected mode for every language model, not evidence that providers allocate identical internal reasoning compute; archival reporting must therefore include the exact model-specific configuration and model snapshot. Three attempts per task capture only limited within-model variability. Task-bootstrap intervals quantify sensitivity to benchmark composition, not verifier uncertainty or future API drift. The present model-call cost excludes host compute.

Domain labels are coarse, task counts are imbalanced, and many instances are synthetic or task-authored. Synthetic construction enables controlled hidden cases but cannot reproduce all noise, ambiguity, and access constraints of experimental science. Strict all-tests pass is transparent but treats every task as one binary unit, so a single formatting failure and a substantive scientific error have the same top-level outcome. If benchmark tasks are made public in the future, task exposure may create contamination over time; benchmark versions and hidden test sets will therefore require maintenance. Furthermore, scientific workflows can touch clinical, genomic, environmental, and dual-use domains. Benchmark outputs must not be interpreted as evidence that an agent is safe for autonomous deployment or that a passed synthetic task validates real-world scientific conclusions.

7 Conclusion and Future Works

ML4SCIENCE-BENCH evaluates whether autonomous agents can complete learned scientific workflows rather than merely generate executable code. Across 3,000 controlled-scaffold runs, strict pass rates range from 24.7% to 77.3%, and per-run model cost varies by more than sixfold. No evaluated model passes every task, and the models with the highest pass rates are not always the fastest or least expensive. These results provide a baseline for measuring progress toward agents that can build scientific ML pipelines whose conclusions survive both computational and scientific tests. Future work could evaluate additional agent scaffolds, use more attempts per task (e.g., reporting Pass@8), and examine different inference budgets. It could also compare model performance across different reasoning settings.

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Table 3: Descriptive audit of the four zero-pass tasks for the top three models. “Failed tests” gives the per-run range across three runs, followed in parentheses by the sum over those runs. Denominators are the tests collected by pytest in each run.

Task	tests/run	Failed tests: range (sum)			Recurrent observed failure families
		Opus 5	GPT-5.6 Sol	Grok 4.6	
ad-inhibitor-ml-replication	27	1–2 (4)	3–5 (12)	2 (6)	Input/path perturbations, target-name aliases, and candidate-target gating
brainstem-rem-gating-dynamics	100	7–10 (24)	3–8 (17)	5–8 (19)	State-space schema, transition/PC2 recomputation, infraslow lags, and cross-artifact contracts
dg-interneuron-wta	43	1–2 (5)	1–2 (4)	2–5 (9)	Spatial tuning, spike-derived features, CCG support, and perturbation propagation
emle-adp-embedding-pipeline	133	2–3 (8)	2–9 (20)	13–104 (134)	Induction/polarization and total EMLE energies, WHAM free-energy surface, and one path-interface cascade

A Failure analysis of the four zero-pass tasks

We inspected the three ML4SCIENCE-BENCH runs from each of the three highest-ranked models, i.e., Opus 5, GPT-5.6 Sol, and Grok 4.6, on the four tasks that received no strict pass from any evaluated model. This gives 36 audited runs. Run directories were matched to the exact time, cost, step, tool-call, and failed-test aggregates used in the paper; for each selected run we read the trial result, executable-test report, and failing assertion trace. All 36 verifiers completed and none of the selected result files records a trial-level exception. The archived reference-solution (oracle) evaluations pass strictly for all four task versions.

Table 3 reports literal failed-test counts. The range is over three runs and the bracketed value is their sum. Counts are not independent failure causes or severity scores: several tests can share one upstream defect, as the EMLE path-interface failure below illustrates.

AD-inhibitor ML replication. The nine runs generally produced the requested pipeline but failed robustness tests. Two Opus 5 runs did not normalize the Unicode β alias in GSK-3 β ; its third run did not propagate mutated resource and argument paths. All three GPT-5.6 Sol runs selected GSK-3B after a perturbation that excluded it from the candidate set and failed to keep publication anchors separate from recomputed data supplied through alternate paths; two also failed the β alias and one failed a reordered/noisy workbook. Two Grok 4.6 runs failed the mutated-resource/path tests, while the third failed workbook robustness and candidate gating. Thus the repeated difficulty was not baseline model fitting alone, but preserving semantic identity and provenance while inputs, paths, and admissible targets changed.

Brainstem REM-gating dynamics. Every audited run failed the state-space-summary test: outputs either used the wrong object structure or an incorrect 90% ellipse scale. Seven of nine runs emitted infraslow lags outside the required ± 10 s window, six reported a PC2 transition accuracy inconsistent with recomputation from their own transition-feature table, and five failed the deterministic bootstrap test. Additional failures involved missing-input audits, artifact-dependency graphs, bounded cross-modal correlations, state occupancy, and optogenetic dose-response schemas. The pattern occurs in all three models (Opus 5: 7–10 failed tests; Sol: 3–8; Grok: 5–8) and shows the difficulty of maintaining a non-local contract across 38 interdependent artifacts rather than producing any one analysis in isolation.

Dentate-gyrus interneuron winner-take-all analysis. Seven of nine runs reported a spatial-information ordering opposite to the task’s recomputed principal-cell/PV comparison. All three Opus 5 runs failed that test, and two also produced a spatial-rate-map figure whose content did not change when the underlying spike/rate-map inputs were perturbed. Two Sol runs failed the spatial comparison; its remaining failures concerned a noncanonical connection label and a dentate-spike latency inconsistent with raw first-spike times. Every Grok 4.6 run failed the test that reported CCG peak statistics track a recomputed asymmetric local-baseline CCG; one run additionally failed canonical cell labels, raw feature reconstruction, a recurrent-circuit ablation relation, and dentate-spike latency. These traces localize the difficulty to propagation from raw spikes and waveforms through classification, connectivity, circuit summaries, and figures.

Table 4: Continuous reward. *Failed-run mean* averages c_{tmr} only over a model’s failed runs; C_m includes all 300 runs. Intervals use the same 20,000 task-bootstrap resamples as the strict-pass analysis. The final column is the difference between continuous reward and strict pass rate in percentage points.

model	Failed runs	Failed-run mean	C_m	95% CI	$C_m - P_m$ (pp)
Opus 5	68	0.930	0.984	[0.977, 0.990]	21.1
GPT-5.6 Sol	81	0.880	0.967	[0.946, 0.983]	23.7
Grok 4.6	84	0.888	0.969	[0.953, 0.982]	24.9
Opus 4.8	106	0.891	0.962	[0.948, 0.974]	31.5
Sonnet 5	131	0.892	0.953	[0.939, 0.966]	39.0
GPT-5.6 Terra	133	0.871	0.943	[0.925, 0.958]	38.6
Gemini 3.7 Flash	155	0.591	0.789	[0.748, 0.827]	30.5
GLM 5.2	160	0.857	0.924	[0.903, 0.943]	45.7
Kimi K3	186	0.860	0.913	[0.890, 0.934]	53.3
Gemini 3.1 Pro	226	0.762	0.821	[0.782, 0.857]	57.4

EMLE adaptive-embedding pipeline. All nine runs exceeded tolerance in at least some induction/polarization energies and consequently in total EMLE energies. Opus 5 was closest by test count (two or three failures per run): only four or five of 144 energy elements missed the array tolerance in the displayed traces, with the largest listed induction/polarization discrepancy between approximately 0.11 and 0.30 kcal/mol. Two Sol runs additionally failed sampled static-energy tests at six windows, while its third run failed only the two aggregate energy arrays. Two Grok 4.6 runs reconstructed substantially different WHAM free-energy surfaces, minima, or RMSE values in addition to the energy-array failures. In the third Grok run, passing a string input directory into code that expected a path object raised a `TypeError`; that single interface defect prevented result construction and cascaded to 104 of 133 tests. The task therefore combines tight numerical tolerances with correct units, energy decomposition, WHAM iteration, and a robust public function interface.

This audit is descriptive rather than a causal taxonomy: it records what the executable tests observed in these 36 runs. It does not estimate how often each failure family would recur under new samples, and it does not treat the number of triggered tests as a measure of scientific severity.

B Continuous-reward details

Ordering and task-level behavior. The continuous and strict model rankings remain strongly related (Spearman $\rho = 0.915$ across ten models), but they are not identical: continuous reward places Grok 4.6 0.13 points above GPT-5.6 Sol. A paired task bootstrap gives Sol minus Grok as -0.13 points (95% CI $[-2.41, 1.84]$), so this small reversal is unresolved. Opus 5 exceeds Sol by 1.67 points (95% CI $[0.23, 3.64]$) and Grok by 1.54 points (95% CI $[0.27, 2.98]$) under the continuous metric. Across tasks, pooling models, strict pass rate and continuous reward have Spearman correlation $\rho = 0.706$, indicating substantial but incomplete agreement. Note that, the wide gaps in the final column of Table 4 are therefore expected: strict pass maps any positive failed-test count to zero, whereas C_m retains all passed tests.

Failed runs often pass a large fraction of executable tests. Under Eq.(3), the mean continuous reward among failed runs ranges from 0.591 to 0.930 across models, while model-level continuous reward ranges from 0.789 to 0.984. Nevertheless, this test-count metric is not calibrated to scientific severity because tests differ in granularity and a single upstream error may trigger multiple downstream failures.

Why continuous reward remains secondary. The four zero-pass tasks illustrate the distinction. Their pooled continuous rewards are 0.766 for `dg-interneuron-wta`, 0.817 for `ad-inhibitor-ml-replication`, 0.876 for `emle-adp-embedding-pipeline`, and 0.914 for `brainstem-rem-gating-dynamics`, despite none producing a complete run. As Appendix Section A shows, a small number of failed tests can encode non-local scientific inconsistencies, tight numerical failures, or an interface defect that invalidates the requested workflow. Conversely, a single upstream error can trigger many downstream tests. Equal test weighting is consequently transparent and reproducible.

C Detailed empirical results

Table 5: Strict pass rate (%) by harmonized primary scientific subject. Rates macro-average task pass fractions; each task has three runs.

Model	Biology	Chemistry	Mathematics	Physics
Opus 5	81.9	77.8	100.0	65.7
GPT-5.6 Sol	72.5	77.8	100.0	65.7
Grok 4.6	74.6	63.9	91.7	66.7
Opus 4.8	73.2	63.9	87.5	48.0
Sonnet 5	61.6	41.7	87.5	47.1
GPT-5.6 Terra	50.7	58.3	87.5	53.9
Gemini 3.7 Flash	47.1	41.7	75.0	46.1
GLM 5.2	50.0	30.6	83.3	39.2
Kimi K3	44.9	30.6	45.8	29.4
Gemini 3.1 Pro	26.1	30.6	33.3	18.6

Table 6: Interaction footprint and strict-pass efficiency with all models in high-thinking mode. Per-pass quantities divide total interactions by strict passes.

Model	Steps/run	Tools/run	Tools/step	Steps/pass	Tools/pass
Opus 5	20.5	27.4	1.34	26.5	35.4
GPT-5.6 Sol	12.3	24.3	1.97	16.9	33.3
Grok 4.6	26.8	28.1	1.05	37.3	39.1
Opus 4.8	23.2	27.1	1.17	35.9	41.9
Sonnet 5	44.7	53.2	1.19	79.4	94.4
GPT-5.6 Terra	13.6	27.5	2.02	24.4	49.5
Gemini 3.7 Flash	56.9	43.3	0.76	117.6	89.6
GLM 5.2	24.1	49.3	2.05	51.6	105.6
Kimi K3	27.5	38.2	1.39	72.4	100.5
Gemini 3.1 Pro	14.0	22.7	1.62	56.7	92.1

Table 7: Tasks at the extremes of observed difficulty, pooled over 10 systems and 30 runs per task.

Task	Subject	Passes	Pass rate
ad-inhibitor-ml-replication	Chemistry	0/30	0.0%
brainstem-rem-gating-dynamics	Biology	0/30	0.0%
dg-interneuron-wta	Biology	0/30	0.0%
emle-adp-embedding-pipeline	Chemistry	0/30	0.0%
adaptive-bci-decoder-simulator	Physics	2/30	6.7%
nanophotonic-classifier-pipeline	Physics	2/30	6.7%
quantum-detector-error-model	Physics	28/30	93.3%
regulome-direct-target-attribution	Biology	28/30	93.3%
biopython-kmer-amplicon-classifier	Biology	29/30	96.7%
conformer-ensemble-analysis	Chemistry	29/30	96.7%
peptide-ensemble-analysis	Chemistry	29/30	96.7%
cr2-qicas-strategy-selector	Physics	30/30	100.0%

The subject table uses a primary scientific subject assigned independently of performance. Entries from the original Data Science and Machine Learning working buckets were reassigned after an instruction-level audit; those terms remain cross-cutting methods rather than scientific subject labels. Every model in these comparisons uses high-thinking mode. Table 6 complements the leaderboard with interaction footprints and per-pass efficiency.

Steps and tool use. Pooling attempts across tasks and models, successful runs average 24.4 trajectory steps and 30.8 tool calls, compared with 28.9 steps and 38.3 calls for failed runs. This marginal comparison is strongly composition-dependent: failures are concentrated on harder tasks and in models with different interaction policies. A within-cell analysis reverses the direction. Among 333 task–model cells (covering 92 tasks) that contain both outcomes, passing attempts use 5.1 more steps on average than failures (task-cluster bootstrap 95% CI [3.3, 7.0]) and 3.2 more tool calls (95% CI [1.4, 5.1]). A plausible interpretation is that some unsuccessful attempts terminate before completing the workflow; the aggregates do not reveal termination causes, so this is not a causal claim.

Table 8: Paired task-bootstrap differences in strict pass rate among the top three systems. Positive values favor the first system.

Comparison	Difference (pp)	95% CI (pp)
Opus 5 – GPT-5.6 Sol	4.3	[-2.7, 11.7]
Opus 5 – Grok 4.6	5.3	[-2.0, 12.3]
GPT-5.6 Sol – Grok 4.6	1.0	[-6.0, 8.0]

Ordering in ledearboard: The paired task-bootstrap differences are 4.3, 5.3, and 1.0 percentage points, respectively, and all three 95% intervals include zero (Table 8). The current experiment therefore does not resolve an ordering among the leading three.

Resource demand is also a property of the task. Across the 100 tasks, strict pass rate has Spearman correlation $\rho = -0.51$ with mean steps (95% task-bootstrap CI $[-0.65, -0.34]$) and $\rho = -0.46$ with mean tool calls (95% CI $[-0.60, -0.29]$): harder tasks tend to elicit more interaction. After subtracting additive task and system means from all 1,000 task-model cells, the descriptive Pearson correlations between pass fraction and resource use are only $r = 0.03$ for steps and $r = -0.02$ for tools. Steps and calls are related ($r = 0.78$) but not interchangeable: Table 6 shows large differences in calls per step and interactions per strict pass. Raw interaction volume is therefore confounded by both task difficulty.

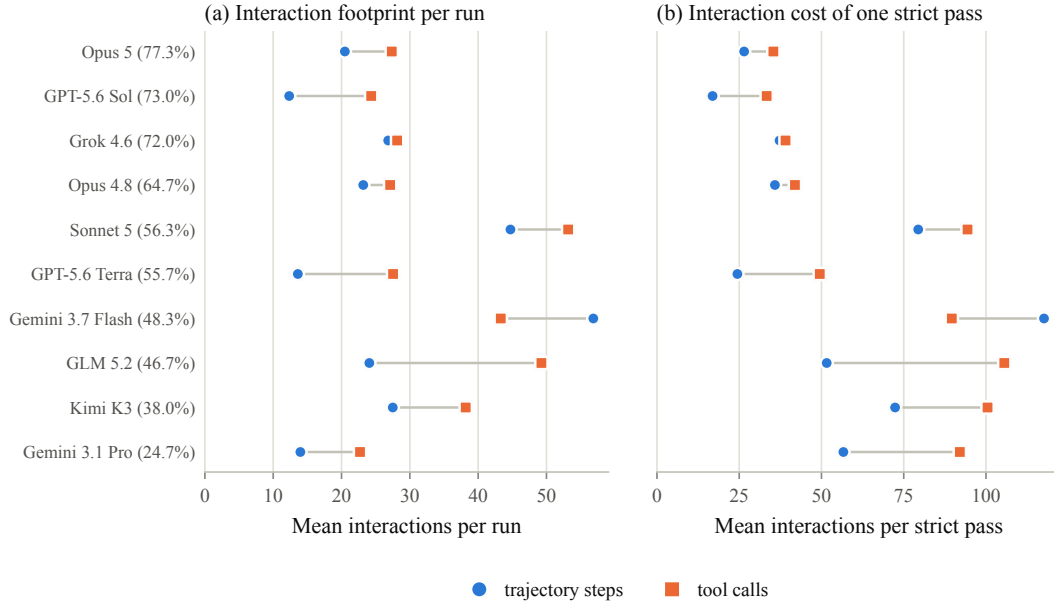


Figure 4: Paired interaction profile by model, ordered by strict pass rate (shown in parentheses). (a) Mean trajectory steps and tool calls per run; (b) the same quantities divided by strict passes, that is, the interaction cost of one success. All agents use the same TERMINUS-2 scaffold in high-thinking mode, so the ordering of the two marks within a row is a tool-use policy difference; steps and calls have different instrumentation semantics and are not interchangeable units.

Subject profile and the granularity of strict pass. Figure 5 plots the subject table graphically and, in panel (b), the distribution over failed runs of the share of a task’s tests that failed. Half of all failed runs fail a tenth or less of the task’s tests, but this does not establish that those failures are minor because test severity and granularity vary across tasks.

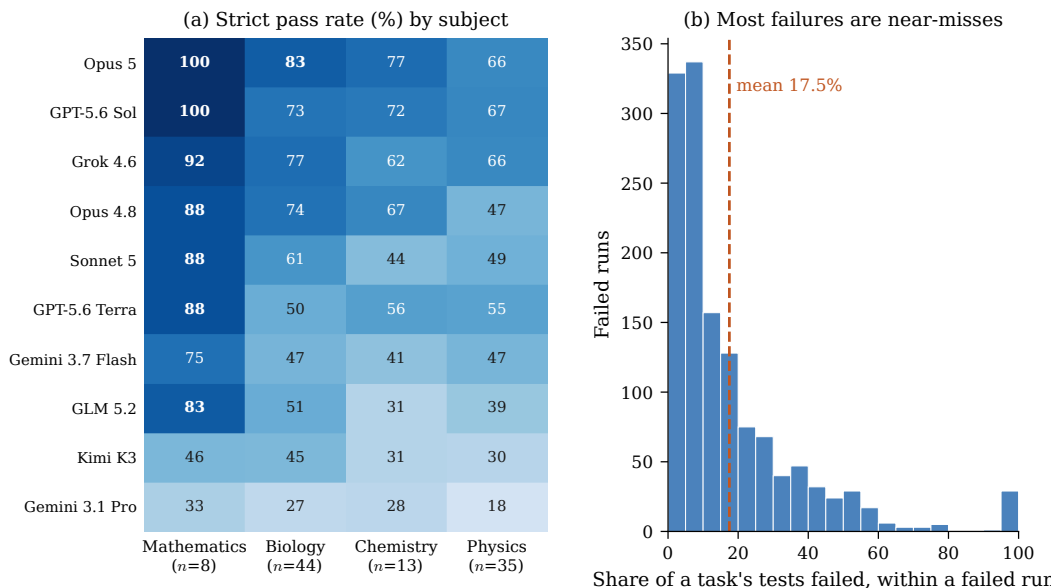


Figure 5: (a) Strict pass rate by primary scientific subject, with the number of tasks per subject; this is the graphical form of Table 5. The subject groups are unbalanced and sometimes interdisciplinary, so the differences are not read as intrinsic domain difficulty. (b) Distribution of the average percentage of tests failed for each run with at least one failed test, covering 1,330 failed runs in total. The bars summarize individual runs. Tasks differ in the number and scope of their tests, and even one failed test may indicate an important scientific error.

D Evaluated task inventory and paper provenance

The evaluated benchmark version contains 100 tasks. Table 9 fixes the empirical denominator by listing exactly the task identifiers used in the evaluation, together with subject, data provenance, pooled pass count, and motivating citation.

Table 9: The 100 evaluated task IDs, primary scientific subject, recorded data provenance, and motivating paper source.

Task identifier	Subject	Data	Runs passed	Source
basal-ganglia-counting	Biology	synthetic	7/30	[32]
biomass-spatial-autocorrelation-audit	Biology	synthetic	19/30	[89]
biophysical-ode-inference	Biology	synthetic	26/30	[27]
biopython-kmer-amplicon-classifier	Biology	synthetic	29/30	[10]
brainstem-rem-gating-dynamics	Biology	synthetic	0/30	[69]
caos-piriform-spleen	Biology	synthetic	15/30	[115]
catalytic-mechanism-discrimination	Chemistry	synthetic	12/30	[123]
cross-cohort-dysbiosis-reproducibility	Biology	synthetic	17/30	[77]
cryoem-reconstruction-attribution	Biology	synthetic	28/30	[28]
cuttag-bias-reconciliation-audit	Biology	synthetic	26/30	[42]
dg-interneuron-wta	Biology	synthetic	0/30	[36]
domain-wall-fmqa-qubo-surrogate	Biology	synthetic	24/30	[50]
erucamide-tm19-retinal	Biology	synthetic	6/30	[111]

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Table 9 continued

Task identifier	Subject	Data	Runs passed	Source
experiment-guided-af3-protein-ensembles	Biology	synthetic	21/30	[72]
grn-causal-edge-attribution	Biology	synthetic	26/30	[51]
hdx-ms-ensemble-structural-support	Biology	synthetic	19/30	[96]
hierarchical-orthogroup	Biology	synthetic	24/30	[73]
hippocampal-theta-goal-vector-sweeps	Biology	synthetic	14/30	[118]
histaminergic-state-memory-accessibility	Biology	synthetic	13/30	[81]
hlca-label-transfer-reconciliation	Biology	synthetic	23/30	[98]
hmv-tau-burst-suppression	Biology	synthetic	15/30	[39]
isoform-singlecell-reconciliation	Biology	synthetic	6/30	[114]
kernelmd-multiome-control	Biology	synthetic	18/30	[88]
left-truncated-frailty-inference	Biology	synthetic	27/30	[3]
live-imaging-event-decomposition-tb	Biology	synthetic	13/30	[78]
ms-csf-ocb-benchmark	Biology	synthetic	17/30	[5]
nmr-amorphous-structure-evidence	Biology	synthetic	26/30	[21]
ood-molecular-source-transfer	Biology	synthetic	21/30	[63]
organoid-metabolic-identity	Biology	synthetic	26/30	[106]
phosphorus-tipping-point-identifiability	Biology	synthetic	5/30	[91]
pleiotropic-gwas-causal-effect	Biology	synthetic	13/30	[100]
pooled-screen-phenotype	Biology	synthetic	5/30	[25]
regulome-direct-target-attribution	Biology	synthetic	28/30	[33]
rna-evolutionary-function	Biology	synthetic	25/30	[34]
saxs-protein-conformation-fitting	Biology	synthetic	18/30	[119]
schic-structural-zero-ranking	Biology	synthetic	22/30	[67]
scrna-seq-patient-stratification	Biology	synthetic	20/30	[37]
seismic-mechanism-attribution	Physics	synthetic	16/30	[92]
selective-del-hits-calibration	Biology	synthetic	16/30	[40]
spatial-crossomics-correction	Biology	synthetic	2/30	[59]
spatial-cv-map	Biology	synthetic	27/30	[89]
spatial-niche-counterfactual-prediction	Biology	synthetic	17/30	[79]
striatal-ach-saliency-circuit	Biology	synthetic	17/30	[16]
striatal-dopamine-trajectory-modeling	Biology	synthetic	22/30	[62]
stt-attractor-role-attribution	Biology	synthetic	17/30	[124]
tcpr-prp-antigen-discovery	Biology	same as paper	16/30	[107]
ad-inhibitor-ml-replication	Chemistry	synthetic	0/30	[4]
band-gap-calibrated-molecular-functional	Chemistry	synthetic	19/30	[128]
conformer-ensemble-analysis	Chemistry	synthetic	29/30	[105]
cox-qeq-reweighting-analysis	Chemistry	synthetic	11/30	[94]
eis-kramers-kronig-relaxation-inference	Chemistry	synthetic	25/30	[109]
emle-adp-embedding-pipeline	Chemistry	synthetic	0/30	[126]
multistate-chemical-md-pipeline	Chemistry	synthetic	21/30	[13]
pde2-active-alchemy-screening	Chemistry	synthetic	9/30	[49]
peptide-ensemble-analysis	Chemistry	synthetic	29/30	[76]
polymer-shear-bmpc	Chemistry	synthetic	8/30	[56]
scaffold-aware-solubility-modeling	Chemistry	synthetic	17/30	[99]

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Table 9 continued

Task identifier	Subject	Data	Runs passed	Source
uncertainty-cv-sampling	Chemistry	synthetic	18/30	[103]
anomalous-transport-discrimination	Mathematics	synthetic	27/30	[82]
gpinn-forward-loss	Mathematics	synthetic	25/30	[120]
mmao-cls-metabolic-generation	Mathematics	synthetic	25/30	[113]
noise-robust-equation-discovery	Mathematics	synthetic	18/30	[71]
otno-geometry-embedding-audit	Mathematics	synthetic	16/30	[61]
stein-variational-bayesian-inversion	Mathematics	synthetic	26/30	[18]
streaming-tt-ice-compression	Mathematics	synthetic	26/30	[2]
xy-model-topological-defects	Mathematics	synthetic	27/30	[86]
adaptive-bci-decoder-simulator	Physics	synthetic	2/30	[68]
bifeo3-urbach-tail-screening	Physics	same as paper	20/30	[6]
bluff-body-interference-xai	Physics	synthetic	25/30	[47]
bounded-aerofoil-rom	Physics	synthetic	14/30	[41]
cacp-solar-fleet-forecasting	Physics	synthetic	27/30	[80]
chalcogenide-catalyst-surrogate	Physics	synthetic	3/30	[35]
cof-photophysics-inverse-solver	Physics	same as paper	9/30	[64]
computational-spectrometer-inverse-reconstruction	Physics	synthetic	7/30	[60]
cr2-qicas-strategy-selector	Physics	same as paper	30/30	[26]
electro-ionic-qa-landscape	Physics	same as paper	15/30	[9]
fatigue-crack-fleet-rul	Physics	synthetic	24/30	[30]
forte-benchmark-audit	Physics	same as paper	10/30	[17]
generalized-qft-resource-sensitivity	Physics	same as paper	18/30	[57]
gkp-qudit-memory	Physics	synthetic	14/30	[12]
gw-relaxation-inverse-analysis	Physics	same as paper	3/30	[52]
heisenberg-quantum-feature	Physics	synthetic	15/30	[93]
hybrid-memristor-accelerator-solver	Physics	synthetic	7/30	[75]
imd_vortex_gust	Physics	synthetic	10/30	[54]
induced-seismicity-synthetic-control	Physics	synthetic	20/30	[112]
majorana-truncation-resource-ranking	Physics	same as paper	5/30	[24]
metamagnetic-endpoint-scaling	Physics	synthetic	13/30	[66]
multilayer-optical-constant-calibration	Physics	synthetic	23/30	[7]
nanophotonic-classifier-pipeline	Physics	synthetic	2/30	[121]
neutrino-hierarchy-phase-reconstruction	Physics	synthetic	24/30	[8]
optical-gkp-homodyne-tomography	Physics	synthetic	15/30	[53]
pcm-drift-short	Physics	synthetic	11/30	[108]
phase-transition-thermometer-inverse-design	Physics	same as paper	7/30	[1]
quantum-detector-error-model	Physics	synthetic	28/30	[116]
quench-slowdown-reconciliation	Physics	synthetic	22/30	[85]
radical-pair-policy-frontier	Physics	same as paper	21/30	[20]
vibronic-gbs-hierarchy	Physics	synthetic	4/30	[31]
viv-strong-coupling-rom	Physics	synthetic	18/30	[125]
wall-turbulence-xai	Physics	synthetic	5/30	[23]
wen16-cvxpy-topological-entropy	Physics	same as paper	19/30	[44]

E Statistical analysis and result integrity

The evaluation matrix contains 100 unique task identifiers, the same 10 models in high-thinking mode on every task, and exactly three attempts in every task–model cell. This balanced design yields 300 runs per model and permits the task-macro pass rate to equal the micro-average over all runs. model-level cost, time, step, tool-call, failure, and reward statistics are recomputed with run-count weights. Automated consistency tests verify that each status-conditioned mean lies between its corresponding recorded minimum and maximum and that all empirical pass counts are integral and bounded by the planned repetitions.

Bootstrap algorithm. For each model we construct a length-100 vector of task pass fractions in $\{0, 1/3, 2/3, 1\}$. A bootstrap replicate samples 100 indices uniformly with replacement and averages the selected fractions. Pairwise model intervals apply the same operation to the length-100 vector of within-task score differences. Interaction-analysis bootstraps also resample tasks: Spearman correlations are recomputed on the sampled task summaries, while matched success–failure comparisons retain every mixed-outcome model cell belonging to each sampled task. We report the 2.5th and 97.5th percentiles across 20,000 replicates. The base extraction seed is 20260824.

F Expanded comparison to related benchmarks

Fixed scientific-ML workflows. PDEBench, Matbench, OC20, and SciML-Bench provide standardized scientific datasets, reference baselines, and evaluation protocols for reproducible model comparison [15, 29, 102, 104]. PDEBench contributes an extensive benchmark of forward and inverse partial differential equations with physics-aware error metrics (RMSE, nRMSE), standing out among non-agentic benchmarks by including parameter inversion [102]. Matbench standardizes 13 materials property prediction tasks across crystallographic datasets, while Open Catalyst 2020 (OC20) delivers millions of density functional theory relaxations for catalyst adsorbate-surface interactions [15, 29]. SciML-Bench provides an execution and benchmarking harness targeting high-performance scientific machine learning systems across multiple natural-science domains [104]. While these benchmarks expose rich scientific structures, they hold the broader computational workflow constant: data partitions, spatial-temporal discretizations, target formulations, and loss functions are predefined. ML4SCIENCE-BENCH asks whether an autonomous agent can construct, calibrate, and audit that modeling contract from scratch.

End-to-end ML engineering and research agents. MLE-bench, FML-bench, MAgentBench, MLGym, and AIRS-Bench evaluate autonomous agents on iterative algorithmic optimization and machine learning engineering over executable repositories [14, 43, 70, 84, 127]. MLE-bench benchmarks end-to-end engineering across 75 Kaggle competitions, scoring submissions against private test leaderboards and medal thresholds [14]. FML-bench supplies executable codebases across eight core machine learning challenges—generalization, sample efficiency, continual learning, representation learning, causality, robustness, privacy, and fairness—measuring exploration diversity and step-level progress alongside predictive gains [127]. MAgentBench and MLGym evaluate language agents on iterative experimentation and AI research progression [43, 84], while AIRS-Bench scales this paradigm to long-horizon frontier AI research challenges derived from published literature [70]. These benchmark excel at testing iterative optimization over codebases; however, they operate on abstract or canonical ML benchmarks devoid of physical conservation laws, domain units, spatial autocorrelation constraints, or mechanistic validity gates. ML4SCIENCE-BENCH adapts this executable environment paradigm to natural-science targets governed by strict domain validity contracts.

Scientific analysis, tool orchestration, and reproducibility agents. ScienceAgentBench and SciAgentGym evaluate authentic scientific programming, data analysis, and multi-step tool orchestration [19, 95]; CORE-Bench and PaperBench target computational reproducibility and research paper replication [97, 101]; AstaBench spans broad scientific research assistance [11]; and COMPOSITE-STEM benchmarks expert-authored STEM reasoning [110]. ScienceAgentBench evaluates 102 paper-derived tasks across four natural sciences, scoring program execution, output validity, and token cost, though many tasks isolate data wrangling subroutines and delegate visual figures to LLM judges [19]. SciAgentGym evaluates multi-step scientific tool use with two complementary metrics: success rate, based on whether all subquestions are answered correctly, and success-weighted path

length, which measures the efficiency of successful tool-use trajectories relative to expert-verified reference paths. [95]. CORE-Bench evaluates computational reproducibility across 270 tasks from 90 papers, requiring agents to resolve legacy dependency conflicts and reproduce published numerical values [97]. PaperBench formalizes end-to-end research replication by tasking agents with rebuilding 20 machine learning papers from scratch, scoring execution against 8,316 hierarchical rubric leaves using an automated judge [101]. AstaBench targets open-ended research assistance—literature retrieval, QA, table generation, and computational reporting—predominantly within computer science workflows [11]. COMPOSITE-STEM pairs 70 expert-curated reasoning and computation problems across STEM disciplines with containerized environments, scoring Pass@1 through exact-match parsing and LLM-Judge rubrics [110]. In contrast, ML4SCIENCE-BENCH enforces a focused scope where every evaluated task centers on assembling, fitting, calibrating, and defending a learned scientific model under deterministic validity gates.

Domain-specific workflows, physical simulation, and discovery benchmarks. BioXArena evaluates executable biomedical machine learning, SciAgentArena benchmarks multi-scale biomedical agent interactions, MDGym assesses molecular dynamics simulation workflows, and DiscoveryBench and NewtonBench target hypothesis and symbolic law discovery [55, 58, 65, 74, 122]. BioXArena evaluates agents across 76 multimodal biomedical tasks spanning nine domains, measuring predictive performance on hidden test splits using domain-specific metrics [58]. SciAgentArena provides an interactive, agent-agnostic framework spanning roughly 200 biomedical challenges across molecular to patient scales, evaluating stepwise verification and feasibility assessment [65]. MDGym requires agents to operate real domain software (LAMMPS and GROMACS) to compute physical observables against reference trajectories within fixed relative tolerances, but contains no learned statistical component [55]. DiscoveryBench and NewtonBench focus on data-driven discovery, evaluating qualitative hypothesis formulation and symbolic equation recovery from observational time series [74, 122]. These benchmarks address distinct facets of scientific computation; ML4SCIENCE-BENCH bridges these spaces by evaluating end-to-end scientific ML pipelines where domain simulators, surrogate modeling, inverse inference, and statistical calibration are tightly coupled.

Spectrum of verifications. Evaluating agent workflows often motivates model-based evaluators, but judge reliability is itself an active research challenge [117]. Prior benchmarks span a clear architectural continuum in how correctness is certified:

- *LLM judge in the primary scoring loop:* PaperBench resolves 8,316 rubric leaves via an automated LLM judge [101], and ScienceAgentBench routes non-programmatic outputs (e.g., visual plots) to a LLM judge with numerical acceptance thresholds [19].
- *Dynamic rubric extraction and scoring:* AstaBench employs an LLM judge to dynamically extract and cluster grading rubrics from long-form research text and score citation support [11].
- *Declared criterion-level Judge partitioning:* COMPOSITE-STEM explicitly assigns each sub-criterion manifest to either exact-match programmatic parsing or an LLM Judge with a specialized prompt, allowing deterministic and Judge tests to coexist [110].
- *Deterministic scoring with post-hoc judge:* MDGym scores reported molecular-simulation quantities against reference values using a 5% relative tolerance; it separately uses rule-based and LLM-based methods to analyze failures. CORE-Bench evaluates reproduced results by requiring every reported answer to fall within a 95% prediction interval derived from three manual reproductions [55, 97]. Aviary, [83], evaluates language agents on multi-step, tool-based scientific tasks in interactive environments, complementing ML4SCIENCE-BENCH’s focus on constructing and deterministically verifying end-to-end scientific machine-learning workflows.
- *Deterministic assertion (ML4SCIENCE-BENCH):* All strict claims regarding numerical precision, data leakage, uncertainty calibration, physical conservation laws, and artifact integrity are evaluated through deterministic programmatic tests ($y_{tmr} = \prod_{k=1}^{K_t} z_{tmrk}$). A failure on any individual scientific assertion is fatal, preventing superficial predictive gains from masking fundamental modeling defects.